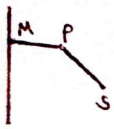
Ellipse



Locus of pt which moves in a plane such that ratio of its distances from a fixed point (S) to its dist. from a fixed line is always a const. & less than 1

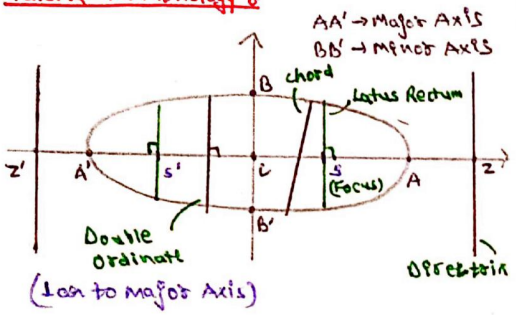
i.e. $\frac{SP}{PM} = e < 1$

Cond^n on 2nd degree Eqn for Ellipse ⇒

(i) $\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} \neq 0$

(ii) $h^2 - ab < 0$

General Terminology :

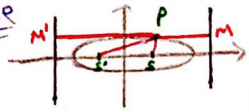


→ Ellipse has two foci & two directrix.

→ P.O.I of Major & Minor axis is centre (C).

→ Ellipse is symmetric about x & y axis.

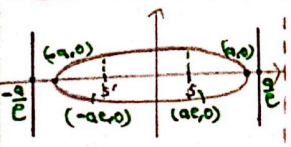
→ While calculating eccentricity, always consider distances from corresponding directrix of focus only.



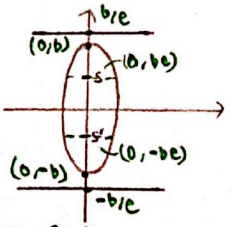
For $\frac{SP}{PM} = \frac{S'P}{PM'} = e$

While $\frac{SP}{PM'} \neq e$

Two Standard Ellipse :



Centre = (0,0)
Vertices = (±a, 0)
Major Axis = 2a
Minor Axis = 2b
Focus = (±ae, 0)
Directrix = $x = \pm \frac{a}{e}$
Relation in a, b, e : $b^2 = a^2(1-e^2)$
Length of LR = $\frac{2b^2}{a}$
Ends of LR = $\left(\pm ae, \pm \frac{b^2}{a}\right)$



Centre = (0,0)
Vertices = (0, ±b)
Major axis = (2b)
Minor axis = (2a)
Focus = (0, ±be)
Directrix = $y = \pm b/e$
Relation in a, b, e : $a^2 = b^2(1-e^2)$
Length of LR = $\frac{2a^2}{b}$
Ends of LR = $\left(\pm \frac{a^2}{b}, \pm be\right)$

Eqn of Standard Ellipse ⇒

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

(a > b) → Horizontal Ellipse

(a < b) → Vertical Ellipse

Note :

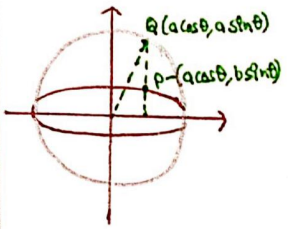
1) Length of LR = $\frac{(\text{Minor Axis})^2}{\text{Major Axis}}$

2) Length of LR = 2e × (Distance b/w focus & corresponding directrix)

= $2e \times \left(\frac{a}{e} - ae\right)$

Auxiliary Circle and Eccentric Angle :

→ A circle concentric with ellipse, with its major axis as diameter is called auxiliary circle.



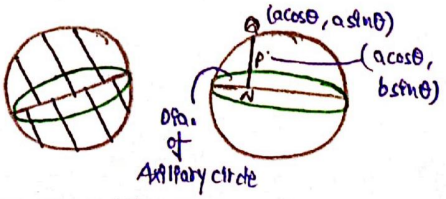
* θ is called Eccentric angle of P

* P and Q are called Corresponding Points on ellipse and auxiliary circle

* If, from each point on a circle, lines are drawn on a fixed diameter then the locus of a point dividing these lines in a const. ratio is an ellipse whose auxiliary circle is the original circle.

We have,

$$\frac{PN}{QN} = \frac{b\sin\theta}{a\sin\theta - b\sin\theta} \Rightarrow \frac{b}{a-b} = \text{const.}$$



Alternate definition of Ellipse :

It is locus of a point P such that sum of its distance from two fixed points (S_1 & S_2) is always a const. (Length of major axis) and more than the distance b/w fixed points



i.e. $PS_1 + PS_2 = 2a > 2ae$

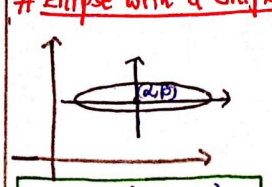
Let, $PS_1 + PS_2 = \lambda$,

$\lambda > S_1S_2$ ⇒ Locus is ellipse

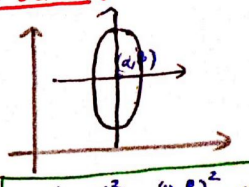
$\lambda = S_1S_2$ ⇒ Locus is line joining S_1 and S_2

$\lambda < S_1S_2$ ⇒ No such point

Ellipse with a shifted centre :

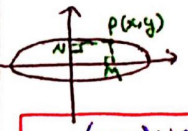


$E : \frac{(x-\alpha)^2}{a^2} + \frac{(y-\beta)^2}{b^2} = 1$ (a > b)



$E : \frac{(x-\alpha)^2}{a^2} + \frac{(y-\beta)^2}{b^2} = 1$ (a < b)

General Version ⇒



$\frac{PN^2}{a^2} + \frac{PM^2}{b^2} = 1$

$$E : \frac{\left(\frac{lx-ly+p}{\sqrt{l^2+m^2}}\right)^2}{a^2} + \frac{\left(\frac{lx+my+n}{\sqrt{l^2+m^2}}\right)^2}{b^2} = 1$$

Position of point w.r.t Ellipse :

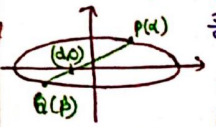
Let the point be $P(x_1, y_1)$, then if

$S_1 < 0$ ⇒ Inside Ellipse

$S_1 = 0$ ⇒ On the Ellipse

$S_1 > 0$ ⇒ Outside the Ellipse

Chord Joining two points P(α) & Q(β) :



$\frac{x}{a}\cos\left(\frac{\alpha+\beta}{2}\right) + \frac{y}{b}\sin\left(\frac{\alpha+\beta}{2}\right) = \cos\left(\frac{\alpha-\beta}{2}\right)$

If this chord passes through (d, 0), we get

$$\tan\left(\frac{\alpha}{2}\right)\cdot\tan\left(\frac{\beta}{2}\right) = \frac{d-a}{d+a}$$

If

(i) d = ae i.e. Focal chord ⇒ $\tan\left(\frac{\alpha}{2}\right)\cdot\tan\left(\frac{\beta}{2}\right) = \frac{e-1}{e+1}$

(ii) d = −ae i.e. Focal chord ⇒ $\tan\left(\frac{\alpha}{2}\right)\cdot\tan\left(\frac{\beta}{2}\right) = \frac{e-1}{e+1}$